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A Systematic Review of Artificial Intelligence, Natural Language Processing, and Explainable Machine Learning Techniques for Clinical Decision Support and Personalized Healthcare Recommendation Systems

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***Abstract*—**Patients are always faced with significant obstacles when attempting to access timely, transparent and affordable healthcare navigation, especially in Tier-2 and Tier-3 cities where centralized information regarding the suitability of clinical specialization and out-of-pocket treatment costs are still significantly constrained. Artificial Intelligence (AI), Natural Language Processing (NLP), and Explainable Machine Learning (XAI) have become important enablers to address these structural information asymmetries, which are key to Clinical Decision Support Systems (CDSS) that manage these constraints. Artificial Intelligence (AI), Natural Language Processing (NLP), and Explainable Machine Learning (XAI) have become important enablers to address these structural information asymmetries, which are key to Clinical Decision Support Systems (CDSS) that manage these constraints. This paper provides a systematic review of forty-six recent works on four domains of AI, namely: clinical decision support systems, medical reasoning and NLP, explainable and trustworthy AI, electronic health record (EHR) representation learning, patient-facing healthcare recommendation systems, security and trust in digital health infrastructure and domain-specific clinical applications. By summarizing the works surveyed with respect to the used methodologies, modalities of clinical data and the operational constraints, they show that there are still some areas of research that are lacking, such as multi-turn clinical ambiguity resolution, integrated cost modeling of medical procedures and the absence of safety rails in automated symptom checkers.

***Index Terms*—**Clinical Decision Support System, Artificial Intelligence, Natural Language Processing, Explainable AI, Healthcare Recommendation System, Multi-Label Classification, Primary Care Triage, Machine Learning, Systematic Literature Review.

I. INTRODUCTION

Today's healthcare environment produces an unparalleled amount of clinical information in an unstructured and semi-structured format, such as clinical encounter notes, diagnostic summaries, imaging reports, pharmacotherapy prescription and patient-reported symptomatology. At the same time, non-expert patients face serious problems trying to find clinical specialisations, and health services that have been verified and predict medical costs before going to a health institution. Most conventional digital health websites serve as a list of facilities and practitioners, with keyword search results that aren't calibrated or personalized and don't include clear treatment costs. This challenge is especially acutely manifested in Tier-2 and Tier-3 cities, where medical records are not consolidated, there is no system of accredited databases and not all the fee schedules are transparent, which leads to delayed diagnosis, unanticipated financial burdens and sub-optimal clinical care pathways.

Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and Explainable AI (XAI) have been extensively studied as pillars of today's Clinical Decision Support Systems (CDSS) that have been designed to address those challenges. The shallow tabular classifiers were succeeded by transformer-based designs in the past 5 years, among which multimodal foundation representations and retrieval-augmented large language model (LLM) agents are able to process complex, longitudinal records and biomedical text. At the same time, the demands for safety, fairness, and regulatory compliance in critical healthcare environments have spurred extensive studies on architectures that lend themselves to interpretation, on post hoc audits, and on strong data protection protocols.

Most crucially, an AI tool that is used in clinical environments where patients interact directly with it is limited to a set of clinical safety protocols: It is not allowed to be an independent diagnostician, nor is it permitted to confirm a medical diagnosis. Reliable systems should be used as initial symptom navigation and clinical triage support systems, which help patients recognize their potential preliminary clinical issues, provide them with additional clinical context through multi-turn follow-ups and guide them to qualified specialist physicians for formal diagnosis, physical examination, and laboratory examination.

The aim of this work is to review in a systematic way 46 peer-reviewed research papers (2019–2026) collected to build the scientific groundwork for our engineering capstone project: MediPath AI: an intelligent clinical triage and verified hospital navigation. This review is unique in its contributions because:

1) **Thematic Taxonomy:** A taxonomy of forty-six recent, peer-reviewed contributions across seven different thematic groups, covering modern healthcare decision support with the use of AI.
2) **Comparative Methodological Analysis:** An analytical comparison that is performed by comparing model architectures, clinical data modalities, evaluation paradigms and structural limitations between the examined systems.
3) **Clinical Research Gaps:** A unified matching of specialists and procedure cost estimation gap in the literature, as well as a lack of multi-turn conversational disambiguation before finalizing the triage, and a lack of explanations that patients can understand, and under-served geographic contexts.
4) **System Architecture and Validation (MediPath AI):** The design, implementation and empirical validation of MediPath AI (MedP_AI_V2.0): a 56-class Multi-Label Primary-Care Triage Engine with mandatory counterquestioning in the 'Turn-0', a 57th GP Safe-Deferral route, deterministic Emergency Overrides and Tiered Hospital/cost routing with exact Mathematical serving parity.

The remaining of this article goes like this. The Methodology for the Review is explained in Section II. The related literature is surveyed, with respect to the seven thematic domains in Section III. Comparative analyses and Platform comparisons are given in Section IV. In the identified research gaps detailed in Section V, research was sought to address the aforementioned gaps. In Section VI, the architecture for MediPath AI, the ML triage pipeline and empirical results for evaluating the benchmark of the AI services are presented. The cross-cutting challenges, ethical safeguards and future enhancements are discussed in Section VII. The paper ends with a summary of the results in Section VIII.

II. REVIEW METHODOLOGY

This review captures the literature using a structured literature aggregation framework, specific to clinical decision support, medical language models and healthcare recommendation systems, and is project scoped.

A. Source Identification and Archive Verification

Identify source of data and check the data in the archives. The literature search and indexing were done within the five major scientific repositories: IEEE Xplore, ACM Digital Library, PubMed / NCBI, ACL Anthology. Clinical decision support, NLP, healthcare recommendation, explainable AI, and treatment cost estimation were the controlled terms used in the construction of search strings, which were then used to query the data. Controlled terms used to construct search strings included: ("clinical decision support" OR "CDSS") AND ("natural language processing" OR "NLP") AND ("healthcare recommendation" OR "hospital recommendation") AND ("explainable AI" OR "XAI") AND ("treatment cost estimation" OR "medical triage"). All product announcements, patents, and entries that did not contain a methodological description were removed by a preliminary screening pass, and were only commercial product announcements. Forty-six publications were retained and systematically archived as primary source PDFs into the project repository and DOIs and bibliographic records were checked for each publication to ensure citation completeness and integrity of 100%.

B. Inclusion and Quality Criteria

Publications were selected following three key requirements: (i) they explicitly contained AI, ML, NLP or XAI as the primary application to clinical triage, biomedical information extraction or healthcare navigation of human clinical patients; (ii) they clearly documented model architecture, the provenance of the data used for model development, and quantitative evaluation metrics; and (iii) they were published in a peer-reviewed journal, conference proceeding or recognized indexed repository between 2019 and 2026.

C. Thematic Coding Taxonomy

The forty-six papers that have been verified were classified into the following non-overlapping thematic areas according to the main technical contribution of the paper: (A) AI-Based Clinical Decision Support Systems, (B) Biomedical and Clinical NLP, (C) Explainable and Trustworthy AI in Healthcare, (D) EHR and Time-Series Representation Learning, (E) Healthcare Recommendation Systems and Patient-Facing Applications, (F) Security, Trust, and System Management in Digital Health, and (G) Domain-Specific Clinical Applications.

III. RELATED WORK

A. AI-Based Clinical Decision Support Systems

Intelligent healthcare triage has as its conceptual basis clinical decision support systems (CDSS). Building on this, Öğdü et al. [5] introduced an adaptive multi-agent LLM system that enables the coordination of the retrieval of vectors from the BioBERT-based vector database with real-time web intelligence to cross-validate the symptom analysis on the MedQA, PubMedQA and MedBullets benchmarks. Lin et al. [2] designed an automated screening CDSS using machine learning, which is based on public health surveillance data, at a population level to achieve early case finding for COPD. Misra et al. [28] proposed an ambulatory, IoT-enabled architecture for on-the-fly monitoring and automated suggestion of in-transit patient routing called i-AVR. The concept of trust propagation in distributed clinical service topologies was thus formalized by Gayen et al. [44] and Nazar et al. [39] provided a requirements-driven framework for directly connecting explainability constraints to software engineering requirements. The works reflect an evolutionary path towards multi-source, collaborative and requirements-driven clinical decision pipelines from static classifiers based on a single encounter.

B. Biomedical and Clinical Natural Language Processing

Patient-generated unstructured text needs to be converted to the structured clinical concepts, and this requires natural language processing. Gu et al. [1] (PubMedBERT) showed that a general-domain model is not the best choice for biomedical domains since it can be outperformed by domain-specific pretraining from scratch on biomedical corpora on a variety of biomedical downstream tasks. The contributors of the SciBERT model [15] proposed to improve the performance of the BERT model on biomedical and scientific NLP tasks by pretraining it on large-scale scientific literature. To further improve the performance of the NER, RE and QA in the biomedical domain, Lee et al. [12] proposed a method called BioBERT, which trains a BERT model using biomedical corpora. Luo et al. [13] created BioMedGPT, a single multimodal generative transformer for chemical, genomic and natural language modalities. Feng et al. [37] developed PhenoBERT, a deep architecture-mapping clinical notes to Human Phenotype Ontology (HPO) concepts with the guidance of an ontology, for clinical phenotyping. Karthikeyan et al. [7] addressed the problems of pre-processing in medical records with a motivation to develop a pipeline for post-correction of OCR output. They developed a masked-language-model (MLM) OCR post-correction pipeline, which substantially improves the character error rate in digitized medical reports. Le et al. [17] introduced representations based on semantics for the task of clinical status detection in the context of a narrative discharge text and Le et al. [29] noticed the improvement of clinical text classification in French under data-scarcity constraints with the use of a mixture-of-experts (MoE) module. For LLM hallucination mitigation, Knowledge graph integration was studied by Murali et al. [30] and for the extraction of clinical information from Q&A bots, Crema et al. [34] investigated. Kulsoom et al. [35] and Zhou et al. [36] conducted comprehensive landscape studies of the NLP applications in radiology report analysis, clinical triage and automatic medical coding, respectively. Finally, Ishikawa et al. [6] put forward a data augmentation approach for detecting adverse drug events using NLP and Durán-Silva et al. [10] presented BATRACIO, an automatic classification system for biomedical publications in the different phases of translational research.

C. Explainable and Trustworthy AI in Healthcare

In clinical settings where automated advice impacts on the health of human beings, model interpretability is key. Allahabadi et al. [8] adapted the European Union's guidelines for AI systems to real-world use with their Z-Inspection protocol, and performed a comprehensive audit of a deployed deep-learning system for evaluating how severely a patient is affected by COVID-19 in their lungs. Ghanvatkar and Rajan [23] examined methodological aspects of explainability, enumerating and defining frameworks for evaluating the usability of the explanation and the practitioner's trust in it, rather than just the mathematical correctness. In clinical oncology, Palkar et al. [20] used SHAP and LIME to create-interpretable prognostic biomarkers for modelling patient survival with gliomas. Muhammad et al. [38] examined randomized tree ensembles as fast, easily interpretable substitutes for "black box" deep networks for clinical diagnosis. Saraswat et al. [25] and Amirian et al. [41] conducted comprehensive surveys of the state-of-the-art on AI requirements for responsible, fair and transparent AI in Healthcare 5.0 and medical image analysis respectively. In psychiatric and cognitive computing, Nag et al. [24] and Msosa et al. [46] surveyed affective computing and cognitive analysis techniques, and created trustworthy data and AI environments for predicting crises in depression respectively.

D. EHR and Time-Series Representation Learning

Electronic Health Records (EHRs) consist of time series that are irregularly sampled, and have a high degree of multidimensionality and missingness. To reconstruct missing clinical values without losing the temporal correlations between clinical values, Qian et al. [14] proposed an attention-based recurrent network, named Conditional Self-Attention Imputation (CSAI). In order to address missingness and privacy issues in observations, Karami et al. [42] proposed a conditional generative adversarial network (GAN) to convert a multivariate time series into a synthetic tensor representation, called TimEHR. Song et al. [43] created a model called TrajGPT taking advantage of autoregressive transformers that have selective recurrent attention to model patient health trajectories over irregular outpatient visits. Wang et al. [19] introduced EHR-HGCN, a representation based on heterogeneous graph CNN for patient, diagnosis and clinical text nodes, for structural EHR representations. Conversely, to combine multiple modalities, Imrie et al. [9] created an automated multimodal ensemble learning framework for seamless combination of tabular, imaging and textual modalities.

E. Healthcare Recommendation Systems and Patient-Facing Applications

The closest analogue of clinical navigation platforms to patient facing recommendation systems are those systems. Alleviate is a mobile recommendation system that Saritha et al. [4] developed for predicting student distress using logistic regression and random forest, identification of healthcare professionals using official medical registers and directing users to healthcare facilities using OpenStreetMap. Zou et al. [16] introduced a model called DAI-Net (Dual Adaptive Interaction Network) that can model complex interactions between drugs and make longitudinal patient histories more informative to produce safe polypharmacy prescriptions. Yue et al. [40] designed an architecture, called SDPR, for recommendations, which explicitly models the process of syndrome differentiation (symptoms to syndromes to therapeutic principles to herbs) in traditional medicine. The idea of consumer review modeling was taken forward by Kangoni et al. [22] who enriched sentiment-based

recommendation pipelines by using LLM-based feature engineering of patient drug reviews.

F. Security, Trust, and System Management in Digital Health

When it comes to health care AI, security and operational lifecycle management is essential for production deployment. Kandasamy et al. [18] studied the cybersecurity threats to healthcare institutions in Asia, created mitigating measures based on NIST standards to secure the sensitive, telemetry-enabled pipelines. Automated induction techniques with the help of LLMs to detect vulnerability patterns in medical software codebases were developed by Deng et al. [27]. Zeydan et al. [33] studied the problems of distributed lifecycle management in cloud environments including versioning, cross-border regulatory compliance, and distributed training. Behara et al. [45] provided an alternative solution for the validation of trust in the use of the permissioned blockchain ledgers, in order to create tamper-evident audit trails of clinical decision histories and medication administration in movement disorder.

G. Domain-Specific Clinical Applications

Special clinical applications demonstrate the application of decision support techniques in specific medical fields. Karboub and Tabaa [11] proposed a framework for optimizing the bed allocation using a survival analysis and treatment trajectory to predict the length of stay in a hospital and the use of the resources. In the field of obstetrics, Kırkgöz et al. [26] have used feature-selection pipelines on the maternal datasets to build interpretable risk-models for the classification of neonatal birth weight. Raina and Jha [31] proposed an interactive healthcare system (I2HS) that combines the use of machine learning for diagnosing patients with scheduling appointments. Mir et al. [32] performed a systematic review of ambient assisted living (AAL) technologies for the specific use in geriatric healthcare for implementing continuous monitoring of elderly people in their home environments, by utilizing IoT sensors and machine learning algorithms.

IV. COMPARATIVE ANALYSIS

Table I is an overview of the literature that was surveyed for the seven thematic areas. The comparative analysis brings to light the following predominant patterns of methods: (i) Transformer-based architectures and large language models have gained wide ground in both biomedical NLP and clinical reasoning; (ii) the interpretability of the models is still largely post-hoc and few systems integrate out-of-the-box interpretability mechanisms directly into the light-weight user-facing architecture that enables patient access, and (iii) patient-facing recommendation systems tend to operate on one of two dimensions, namely prescribing pharmacotherapies or routing patients to a physician based on their specialty, rather than meaningfully connecting the two and offering differential level of confidence in the recommendations, and (iv) available commercial directories offer either unverified specialization matching algorithms or have no mechanism to provide differential level of confidence, while generic, online chatbot-based symptom checkers try to make a diagnosis based on their recommended treatments without providing follow-up context.

TABLE I: COMPARATIVE SUMMARY OF REVIEWED LITERATURE BY THEMATIC CATEGORY

Thematic Category	Representative Works	Primary Methodology	Clinical Data Modality	Operational Limitation
CDSS (§III-A)	[2, 3, 5, 28, 39, 44]	Neural embeddings, multi-agent LLMs, trust networks	Tabular EHR, text narratives, streaming vitals	Rarely models procedural cost or out-of-pocket expenditure
Biomedical NLP (§III-B)	[1, 6, 7, 10, 12, 13, 15, 17, 21, 29, 30, 34-37]	Domain pretrained LMs (PubMedBERT, SciBERT, BioBERT), MoE, Graph NLP	Clinical discharge notes, radiology reports, PubMed text	Predominantly English-centric; heavy serving latency in client environments
Explainable AI (§III-C)	[8, 20, 23-25, 38, 41, 46]	SHAP, LIME, Z-Inspection, randomized ensembles	Radiological imaging, oncology tabular risk profiles	Evaluations focus on clinicians; usability for non-expert patients is rarely tested
EHR Time-Series (§III-D)	[9, 14, 19, 42, 43]	Self-attention imputation, conditional GANs, TrajGPT, EHR-HGCN	Longitudinal multivariate EHR sequences	High computational training overhead; complex institutional deployment requirements
Recommendation (§III-E)	[4, 16, 22, 40]	Tree classifiers, interaction networks, syndrome differentiation	User profiles, prescription logs, online reviews	Lacks unified clinical triage and multi-tier procedural cost guidance
Security & Trust (§III-F)	[18, 27, 33, 45]	NIST vulnerability audits, blockchain ledgers, cloud MLOps	System logs, software source artifacts, medication ledgers	Security evaluated largely in isolation from patient clinical outcomes
Domain Apps (§III-G)	[11, 26, 31, 32]	Survival analysis, feature selection, integrated portals	Obstetric records, bed occupancy logs, smart-home sensors	Narrow specialty boundaries; limited generalizability across broad primary care

V. RESEARCH GAPS

Based on an analysis of the literature reviewed, the following four research gaps are crucial for motivating the engineering of MediPath AI:

1) **Disconnection of Clinical Routing from Procedural Cost Transparency:** Existing clinical recommendation systems either target only specialists or doctor's recommendation [4, 16, 40] or only the hospital operational costs and bed occupancy from the administrative point of view [11]. There is no reviewed system to connect clinical specialization triage with realistic, tiered treatment cost estimation among regional healthcare facilities, that link patient facing clinical specialization triage with realistic, tiered treatment cost estimation.
2) **Single-Turn Premature Closure and Absence of Multi-Turn Follow-Up:** Most symptom tools that are built for patients try to classify the condition in the first turn based on the initial free text. Patients often don't report important co-occurring symptoms when making a clinical triage appointment. There are no mandatory discriminative counter questioning steps in systems that lead to a conclusion about historically ambiguous differentials (e.g. Otitis Externa vs. Otitis Media, Anxiety vs. Panic Disorder). Another problem that many current medical chatbots and classifiers have is an inability to avoid making blanket predictions for out-of-distribution data.
3) **Lack of Non-Diagnostic Safety Rails and Safe Deferral Mechanisms:** The absence of Non-Diagnostic Safety Rails and Safe Deferral Mechanisms. Reviewed patient tools fail to contain deterministic override options for acute red flag conditions, malignancy referral baskets and explicit safe deferral options including diversion of vague presentations to General Medicine and not hallucinating an acute specialty.
4) **Underserved Regional Healthcare Contexts and Heavy Runtime Overheads:** For Tier-2/Tier-3 healthcare contexts, that are typically low bandwidth, low resource environments, state-of-the-art transformer pipelines (e.g., 7B–70B parameter LLMs) suffer from high latency, cost and GPU requirements. The serving architecture with bit-level probability parity on lightweight web infrastructure is severely lacking with a lack of such architectures that support zero-Python and microsecond latency.

VI. SYSTEM ARCHITECTURE & EVALUATION: MEDIPATH AI

In response to the research gaps identified, we designed, implemented and empirically tested MediPath AI (Model MedP_AI_V2.0), a complete clinical decision support and hospital recommendation platform.

A. Clinical Purpose and Strict Safety Boundaries

MediPath AI is not a medical diagnostic device, but rather is specifically designed to be used as a clinical routing and symptom navigation assistant. The system is clinically robust to guarantee patient safety:

- **Non-Diagnostic Terminology:**All the model outputs are provided with an explicit terminology, namely 'preliminary possible causes' and 'model statistical probabilities. Definitive clinical diagnostic certainty (DDCC) is not allowed in any interface.
- **Deterministic Emergency Red-Flag Interceptor:** When inputs are associated with acute life-threatening signs (e.g., acute crushing chest pain, unilateral facial drooping/stroke indicators, severe respiratory distress, anaphylaxis) the inputs immediately trigger deterministic rule-based emergency overrides, bypassing normal ML inference, and providing immediate emergency care navigation.
- **Oncology and Complex Malignancy Guard:** Inputs containing the terms 'neoplasm', 'tumor' or 'potential malignancy' are safely blocked and referred for further investigation by the oncology specialist without synthetic probability scoring.
- **57th Fallback Route (Unsupported / Uncertain Condition):** If the symptoms are vague, ambiguous or are out-of-distribution and do not surpass the calibrated class thresholds, the symptoms are routed to a 57th Fallback Route, which safely routes the patient to a General Practitioner, while not forcing a classification that is not supported or is uncertain.

B. System Workflow & Multi-Turn Assessment State Engine

MediPath AI performs an explicit, bi-directional assessment state on the frontend and the Node.js backend, whereas typical single-turn symptom checkers only focus on the front end. The workflow:

1. **Symptom Intake & Normalization:** The user types symptoms in free text, in natural language. Gemini 3.8 Flash (with offline deterministic regex fallback) translates multilingually, normalizes medical entities.
2. **Mandatory Turn-0 Discriminative Follow-Up:** Regardless of the underlying triage model's statistical confidence (99%+), the system always follows up with a discriminative follow up at Turn-0. Includes at least one clinically relevant counter-question that is answered with patient friendly symptom chips (e.g., Otitis Externa vs Otitis Media - ear canal discharge vs deep/ throbbing pain with fever).
3. **Dynamic Evidence Accumulation:** Symptoms are given in the form of chips that the patient can choose to add or remove from, or the patient can "No more symptoms". The system is able to record the confirmedSymptoms, rejectedSymptoms, and askedQuestions, and continuously adjust the differential ranking without re-asking questions.
4. **Clinical Status Triage:** When the user is confident or the user terminates the engine assigns an explicit Clinical Status: possible_cause_ready, safe defer (deferral to GP) or emergency override.
5. **Verified Hospital Routing:** Determines the set of candidate hospitals in the search radius, making a strong distinction between Verified Specialist Facilities (those accredited for the target department) and General-Care Fallback Facilities.
6. **Procedural Cost Estimation:** Out of pocket itemized cost estimates that take into consideration hospital tier (Budget, Mid-Tier, Premium) and NABH accreditation.

C. Machine Learning Triage Engine (MedP_AI_V2.0)

MedP_AI_V2.0 is designed to overcome this flaw in previous one-label classifiers by employing multi-label classification (One-vs-Rest) over 56 canonical primary-care classes.

- **Dataset Curation & Exclusions:** 104,291 curated patient records were taken from 157,036 raw, conversational patient cases. The detailed audit process removed 52,745 chronic conditions, surgical conditions, subspecialty conditions and tertiary conditions (such as heart failure, multiple sclerosis and spinal stenosis), and referred them for a formal physician workup.
- **Leakage-Free Group Partitioning:** In order to avoid data leakage, the partitioning of the records was done strictly by canonical symptom signatures (unique feature groups). Any combination of symptoms that occurred in all the cases was restricted to the same split. The Stratification resulted in 72,741 training records, 15,786 validation records, and 15,764 untouched test records and achieved ≥50 positive instances per class in all splits, and 0.00% signature leakage at all splits.
- **Model Selection & Zero-Python Serving Rationale:** Calibrated LinearSVC had a slightly better validation macro F1 score of 0.6763 vs. 0.6226 for LogisticRegression_OvR which gives independent sigmoid probabilities, P = 1 / (1 + e^-z), with closed form equations. The linear weight matrix (W, b) was directly exported to an in-memory, node.js serving engine. It ensures an exact bit-level equality to the scikit-learn version ($\Delta \leq$ 3.88e-16 on 117 benchmark verification queries) with sub-millisecond serving time, no Python runtime dependency, and no need to install the package.
- **Class-Specific Threshold Calibration:** Instead of the standard 0.50 cutoffs, class-specific thresholds τ_k were optimized per class on the validation set to maximize the individual F1 score in order to get the mean optimal threshold of $\tau = 0.85$ (optimal range [0.79, 0.85]).

D. Hospital Recommendation and Procedural Cost Estimation

The hospital suggestions engine combines verified local facilities (920 facilities, curated) with real-time data from the Google Places API, safeguarded by automatic quota control and fallback on OSM. Geographic proximity is calculated using Haversine formula and Indian city aliases (e.g. Bengaluru/Bangalore, Gurugram/Gurgaon) are resolved. Candidates are allotted to the following clinical tiers for the facilities: Primary Specialization Match (Tier 1), Direct Procedure Match (Tier 2), General-Care Fallback (Tier 3 – initial stabilization) and Regional Referral Hospital (Tier 4).

Cost estimation models were developed for OOHCE across the healthcare facilities in India. The 56 conditions have their normal cost characteristic profiles, including average length of stay (length of days), daily bed charges, diagnostic work up charges and pharmacotherapy costs. The multipliers used for hospital facilities are: Budget (0.60×), Mid-Tier (1.00×) and Premium (1.75×) with a +5% quality adjustment for facilities that are National Accreditation Board for Hospitals & Healthcare Providers (NABH) accredited. This yields minimum and maximum prices for each item, providing a useful financial transparency before admission to a patient.

E. Empirical Benchmark Evaluation

A completely untouched held-out test partition (N = 15,764 records) was used for the evaluation of MedP_AI_V2.0. The model showed good clinical rank and calibration in all multi-label evaluation metrics as shown in Table II.

TABLE II: HELD-OUT TEST EVALUATION (N=15,764)

Evaluation Metric	Test Score
Top-1 / Top-3 / Top-5 Hit Rate	88.19% / 99.03% / 99.80%
Micro Precision / Recall / F1	0.6922 / 0.9032 / 0.7837
Macro Precision / Recall / F1	0.6241 / 0.8796 / 0.7199
Weighted F1 Score	0.7959
Label Ranking Avg Precision (LRAP)	0.9160
Hamming Loss / Brier Score / ECE	0.01329 / 0.01827 / 0.0282

Probability calibration was verified using a Mean Brier Score of 0.01827 and an Expected Calibration Error (ECE) of 0.0282 indicating that model statistical probabilities are a good match for the likelihoods of the canonical symptoms. Top-performing condition categories included Depressive Symptoms (F1: 0.9394, Precision: 0.9231, Recall: 0.9564), Osteoarthritis (F1: 0.9387), Endometriosis (F1: 0.9038), Otitis Media (F1: 0.9016), and Pneumonia (F1: 0.8983). Common clinical overlaps between the two sets of confusion cases were identified by analyzing the frequently confused class pairs – including Pelvic Inflammatory Disease vs. Uterine Fibroids (26 cases) and Common Cold vs. Sinusitis (23 cases) – which were found to be well matched with the clinical relevance of the confusion.Known clinical overlaps between the two sets of confusion pairs were identified by analyzing the frequently confused class pairs, such as Pelvic Inflammatory Disease vs. Uterine Fibroids (26 cases) and Common Cold vs. Sinusitis (23 cases), which was consistent with the clinical necessity of MediPath AI's multi-turn counter-questioning engine to resolve the symptom ambiguities.

VII. DISCUSSION & FUTURE DIRECTIONS

A. Practical Implications & Engineering Trade-Offs

The empirical results demonstrate the importance of the balance between complex deep neural networks and lightweight and interpretable linear models. The latency, potential for hallucination, and infrastructure costs of using LLM systems as triage in patient interactions with the system are potentially non-deterministic, problematic, and expensive, respectively [5, 13]. MediPath AI is decoupled from primary-care classification, which is handled by closed-form logistic regression with in-memory matrix serving, and serves the conversational multilingual normalization task with the same determinism and sub-millisecond latency as the primary-care classification, on top of the zero numerical serving divergence on commodity server hardware provided by the latter.

B. Ethical, Legal, and Privacy Considerations

Implementing PHAWS is complex, and there are ethical considerations to take into account. A wrong hospital or cost

suggestion could lead to a deterioration in the clinical course or to an unnecessary financial burden. To overcome these concerns, MediPath AI clearly informs clients that their estimates are "budgetary" ranges, not hard quotes. Moreover, there is no medical data in plain text on the client's disk by default while processing user symptom narratives and providing consultation assessments, which ensures strict adherence to the Indian Digital Personal Data Protection (DPDP) Act and HIPAA requirements.

C. Future Architectural Roadmap

Opt-in Encrypted Assessment History: addition of client-side AES-GCM-256 encryption of Assessment History in Firebase Firestore to comply with DPDP 'Right to Erasure' requirements; User-Controlled Cryptographic Erasure: configurable session TTL retention to meet DPDP 'Right to Erasure' requirements; Anonymous Population-Level Healthcare Navigation Research: addition of ε-differential privacy noise injection; Automated In-Person Appointment Preparation Checklists: printable doctor visit summaries and Ayushman Bharat / insurance verification guides; and Hospital Bookmarking with side-by-side comparative views for bed capacity, NABH accreditation, and procedural cost tiers.

D. Limitations of This Review

There are a number of constraints on this systematic review. Literature identification included big peer-reviewed databases (IEEE, ACM, PubMed, ACL) and limited the search to the English language, from 2019 to 2026. The empirical comparison of non-indexed grey literature and proprietary commercial algorithms did not include these because they did not disclose their methodology.

VIII. CONCLUSION

In this work, we performed systematic review of forty-six recent research works from four areas including clinical decision support, biomedical natural language processing, explainable AI, and electronic health records, patient-facing recommendation systems, and digital health security. We found three key issues in patient facing triage that needed more research: specialist recommendation in the facility and lack of transparency around treatment costs, danger of premature closure in one-turn symptom checkers, and the lack of non-diagnostic safety rails. To overcome these limitations, we designed, implemented and empirically validated our new AI system, namely MediPath AI (MedP_AI_V2.0). The combination of a calibrated multi-label triage engine across 56 canonical classes (88.19% Top 1 hit rate, 99.80% Top 5 hit rate) along with mandatory Turn-0 counterquestioning, 57th GP safe-deferral class and deterministic emergency overrides with zero-Python Node.js serving parity ($\Delta \leq 3.88e\text{-}16$) creates an effective, robust, and privacy-preserving healthcare navigation solution for under-served communities.

REFERENCES

[1] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, "Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing," ACM Transactions on Computing for Healthcare, vol. 3, no. 1, pp. 1–23, 2021, doi: 10.1145/3458754.

[2] X. Lin, Y. Lei, J. Chen, Z. Xing, T. Yang, Q. Wang, and C. Wang, "A Case-Finding Clinical Decision Support System to Identify Subjects with Chronic Obstructive Pulmonary Disease Based on Public Health Data," Tsinghua Science and Technology, vol. 28, no. 3, pp. 525–540, June 2023, doi: 10.26599/TST.2022.9010023.

[3] C. Comito, D. Falcone, and A. Forestiero, "AI-Driven Clinical Decision Support: Enhancing Disease Diagnosis Exploiting Patient Similarity," IEEE Access, vol. 10, pp. 6878–6888, 2022, doi: 10.1109/ACCESS.2022.3141756.

[4] S. A. N., G. Kalnoor, P. R. Bhat, P. A. Rao, P. Prajwal, and P. T. Pradeep, "Alleviate: An AI-Powered Mobile Application for Personalized Healthcare Management and Doctor Recommendation," IEEE Access, vol. 13, pp. 1357–1372, 2025, doi: 10.1109/ACCESS.2025.3647089.

[5] Ç. U. Öğdü, K. Arslanoğlu, and M. Karaköse, "An Adaptive Multi-Agent LLM-Based Clinical Decision Support System Integrating Biomedical RAG and Web Intelligence," IEEE Access, vol. 13, pp. 11024–11039, 2025, doi: 10.1109/ACCESS.2025.3526842.

[6] T. Ishikawa, T. Yakoh, and H. Urushihara, "An NLP-Inspired Data Augmentation Method for Adverse Event Prediction Using an Imbalanced Healthcare Dataset," IEEE Access, vol. 10, pp. 24879–24889, 2022, doi: 10.1109/ACCESS.2022.3155793.

[7] S. Karthikeyan, A. Garcia Seco de Herrera, F. Doctor, and A. Mirza, "An OCR Post-Correction Approach Using Deep Learning for Processing Medical Reports," IEEE Transactions on Circuits and Systems for Video Technology, vol. 32, no. 5, pp. 2574–2581, May 2022, doi: 10.1109/TCSVT.2021.3096055.

[8] H. Allahabadi, J. Amann, I. Balot, A. Beretta, C. Binkley, J. Bozenhard, R. V. Zicari, et al., "Assessing Trustworthy AI in Times of COVID-19: Deep Learning for Predicting a Multiregional Score Conveying the Degree of Lung Compromise in COVID-19 Patients," IEEE Transactions on Technology and Society, vol. 3, no. 4, pp. 272–289, Dec. 2022, doi: 10.1109/TTS.2022.3195114.

[9] F. Imrie, S. Denner, L. S. Brunschwig, K. Maier-Hein, and M. van der Schaar, "Automated Ensemble Multimodal Machine Learning for Healthcare," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 6, pp. 4213–4224, June 2025, doi: 10.1109/JBHI.2025.3530156.

[10] N. Durán-Silva, J. Carrillo-de-Albornoz, L. Plaza, S. Ricardo, F. A. Massucci, S. Veiga, and A. Ramos-Prats, "BATRACIO: BAsic, TRAnslational, Clinical, research phase identification in bIOmedical publications," IEEE Access, vol. 12, pp. 29707–29723, 2024, doi: 10.1109/ACCESS.2024.3368291.

[11] K. Karboub and M. Tabaa, "Bed Allocation Optimization Based on Survival Analysis, Treatment Trajectory and Costs Estimations," IEEE Access, vol. 11, pp. 114674–114690, 2023, doi: 10.1109/ACCESS.2023.3324687.

[12] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, vol. 36, no. 4, pp. 1234–1240, Feb. 2020, doi: 10.1093/bioinformatics/btz682.

[13] Y. Luo, J. Zhang, S. Fan, K. Yang, Y. Wu, M. Qiao, and Z. Nie, "BioMedGPT: Open Multimodal Generative Pre-trained Transformer for BioMedicine," arXiv preprint arXiv:2308.09442, 2023.

[14] L. Qian, J. A. Raj, H. Logan-Ellis, A. Zhang, Y. Zhang, T. Wang, R. J. Dobson, and Z. Ibrahim, "CSAI: Conditional Self-Attention Imputation for Healthcare Time-Series," IEEE Journal of Biomedical and Health Informatics, vol. 30, no. 7, pp. 5775–5786, July 2026, doi: 10.1109/JBHI.2025.3648181.

[15] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A Pretrained Language Model for Scientific Text," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, Nov. 2019, pp. 3615–3620.

[16] X. Zou, X. He, X. Zheng, W. Zhang, J. Chen, and C. Tang, "DAI-Net: Dual Adaptive Interaction Network for Coordinated Medication Recommendation," IEEE Journal of Biomedical and Health Informatics,

vol. 28, no. 10, pp. 6201–6212, Oct. 2024, doi: 10.1109/JBHI.2024.3425833.

[17] T.-D. Le, R. Noumeir, J. Rambaud, G. Sans, and P. Jouvet, "Detecting of a Patient's Condition From Clinical Narratives Using Natural Language Representation," IEEE Access, vol. 10, pp. 48805–48817, 2022, doi: 10.1109/ACCESS.2022.3172081.

[18] K. Kandasamy, S. Srinivas, K. Achuthan, and V. P. Rangan, "Digital Healthcare - Cyberattacks in Asian Organizations: An Analysis of Vulnerabilities, Risks, NIST Perspectives, and Recommendations," IEEE Access, vol. 10, pp. 11097–11116, 2022, doi: 10.1109/ACCESS.2022.3145372.

[19] G. Wang, X. Lou, F. Guo, D. Kwok, and C. Cao, "EHR-HGCN: An Enhanced Hybrid Approach for Text Classification Using Heterogeneous Graph Convolution Networks in Electronic Health Records," IEEE Journal of Biomedical and Health Informatics, vol. 28, no. 3, pp. 1668–1679, March 2024, doi: 10.1109/JBHI.2023.3346210.

[20] A. Palkar, C. C. Dias, K. Chadaga, and N. Sampathila, "Empowering Glioma Prognosis With Transparent Machine Learning and Interpretative Insights Using Explainable AI," IEEE Access, vol. 12, pp. 32745–32761, 2024, doi: 10.1109/ACCESS.2024.3370238.

[21] W. Jaziri and N. Sassi, "Enhancing Semantic Inference in Healthcare Systems With Ontology-Enriched NLP," IEEE Access, vol. 12, pp. 63821–63836, 2024, doi: 10.1109/ACCESS.2024.3396821.

[22] S. M. Kangoni, O. T. Tshipata, P. S. Nzakuna, V. Paciello, and J.-G. M. Mboma, "Enhancing Sentiment-Driven Recommender Systems With LLM-Based Feature Engineering: A Case Study in Drug Review Analysis," IEEE Access, vol. 13, pp. 121523–121538, 2025, doi: 10.1109/ACCESS.2025.3590326.

[23] S. Ghanvatkar and V. Rajan, "Evaluating Explanations From AI Algorithms for Clinical Decision-Making: A Social Science-Based Approach," IEEE Transactions on Technology and Society, vol. 4, no. 3, pp. 238–249, Sept. 2023, doi: 10.1109/TTS.2023.3289052.

[24] P. K. Nag, A. Bhagat, and R. V. Priya, "Expanding AI's Role in Healthcare Applications: A Systematic Review of Emotional and Cognitive Analysis Techniques," IEEE Access, vol. 13, pp. 68420–68441, 2025, doi: 10.1109/ACCESS.2025.3562131.

[25] D. Saraswat, P. Bhattacharya, A. Verma, V. K. Prasad, S. Tanwar, G. Sharma, and P. N. Bokoro, "Explainable AI for Healthcare 5.0: Opportunities and Challenges," IEEE Access, vol. 10, pp. 84486–84517, 2022, doi: 10.1109/ACCESS.2022.3197671.

[26] H. Kırkgöz, M. C. Nacar, and O. Kurt, "Feature Selection-Based Machine Learning Framework for Birth Weight Prediction: Toward Intelligent Clinical Decision Systems," IEEE Access, vol. 13, pp. 28914–28929, 2025, doi: 10.1109/ACCESS.2025.3538411.

[27] L. Deng, H. Lei, F. Khan, and M. Haque, "GPT-Based Automated Induction: Vulnerability Detection in Medical Software," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 2, pp. 1120–1132, Feb. 2025, doi: 10.1109/JBHI.2024.3481234.

[28] S. Misra, S. Pal, N. Pathak, P. K. Deb, A. Mukherjee, and A. Roy, "i-AVR: IoT-Based Ambulatory Vitals Monitoring and Recommender System," IEEE Internet of Things Journal, vol. 10, no. 12, pp. 10318–10327, 15 June 2023, doi: 10.1109/JIOT.2023.3238116.

[29] T.-D. Le, P. Jouvet, and R. Noumeir, "Improving Transformer Performance for French Clinical Notes Classification Using Mixture of Experts on a Limited Dataset," IEEE Access, vol. 11, pp. 84129–84141, 2023, doi: 10.1109/ACCESS.2023.3302618.

[30] L. Murali, G. Gopakumar, D. M. Viswanathan, R. Raman, and P. Nedungadi, "Integrating LLMs and Knowledge Graphs for Medical AI: Advances, Challenges, and Future Directions," IEEE Journal of Biomedical and Health Informatics, vol. 28, no. 9, pp. 5560–5575, Sept. 2024, doi: 10.1109/JBHI.2024.3409182.

[31] R. Raina and R. K. Jha, "Intelligent and Interactive Healthcare System (I2HS) Using Machine Learning," IEEE Access, vol. 10, pp. 86321–86337, 2022, doi: 10.1109/ACCESS.2022.3197878.

[32] A. A. Mir, A. S. Khalid, S. Musa, M. F. A. Fauzi, N. N. A. Razak, and T. B. Tang, "Machine Learning in Ambient Assisted Living for Enhanced Elderly Healthcare: A Systematic Literature Review," IEEE Access, vol. 13, pp. 106421–106443, 2025, doi: 10.1109/ACCESS.2025.3580961.

[33] E. Zeydan, S. S. Arslan, and M. Liyanage, "Managing Distributed Machine Learning Lifecycle for Healthcare Data in the Cloud," IEEE Access, vol. 12, pp. 114812–114833, 2024, doi: 10.1109/ACCESS.2024.3443520.

[34] C. Crema, F. Verde, P. Tiraboschi, C. Marra, A. Arighi, S. Fostinelli, G. M. Giuffré, et al., "Medical Information Extraction With NLP-Powered QABots: A Real-World Scenario," IEEE Journal of Biomedical and Health Informatics, vol. 28, no. 6, pp. 3672–3683, June 2024, doi: 10.1109/JBHI.2024.3378901.

[35] U. Kulsoom, F. G. Glavin, and M. Bendechache, "Natural Language Processing and Machine Learning for Analysis of Radiology Reports: A Systematic Review," IEEE Access, vol. 13, pp. 14210–14234, 2025, doi: 10.1109/ACCESS.2025.3529810.

[36] B. Zhou, G. Yang, Z. Shi, and S. Ma, "Natural Language Processing for Smart Healthcare," IEEE Reviews in Biomedical Engineering, vol. 17, pp. 4–18, 2024, doi: 10.1109/RBME.2023.3301290.

[37] Y. Feng, L. Qi, and W. Tian, "PhenoBERT: A Combined Deep Learning Method for Automated Recognition of Human Phenotype Ontology," IEEE/ACM Transactions on Computational Biology and Bioinformatics, vol. 20, no. 5, pp. 3120–3131, Sept.-Oct. 2023, doi: 10.1109/TCBB.2023.3276541.

[38] D. Muhammad, I. Ahmed, M. O. Ahmad, and M. Bendechache, "Randomized Explainable Machine Learning Models for Efficient Medical Diagnosis," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 9, pp. 6474–6485, Sept. 2025, doi: 10.1109/JBHI.2024.3491593.

[39] M. Nazar, S. Unar, A. Ahmed, M. M. Su'ud, M. M. Alam, and A. Rahmat, "Requirements-Driven Explainable Artificial Intelligence Framework for Secure and Transparent Clinical Decision Support Systems," IEEE Access, vol. 14, pp. 18230–18249, 2026, doi: 10.1109/ACCESS.2026.3531204.

[40] W. Yue, W. Ji, X. Wang, X. Ma, P. Wang, and X. Wang, "SDPR: Prescription Recommendation With Syndrome Differentiation in Traditional Chinese Medicine," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 5, pp. 3736–3747, May 2025, doi: 10.1109/JBHI.2025.3525507.

[41] S. Amirian, F. Gao, N. Littlefield, J. H. Hill, A. J. Yates Jr., J. F. Plate, L. Pantanowitz, and H. H. Rashidi, "State-of-the-Art in Responsible, Explainable, and Fair AI for Medical Image Analysis," IEEE Access, vol. 13, pp. 58120–58145, 2025, doi: 10.1109/ACCESS.2025.3555543.

[42] H. Karami, M.-A. Hartley, D. Atienza, and A. Ionescu, "TimEHR: Image-Based Time Series Generation for Electronic Health Records," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 12, pp. 9170–9181, Dec. 2025, doi: 10.1109/JBHI.2025.3577328.

[43] Z. Song, Q. Lu, H. Zhu, D. Buckeridge, and Y. Li, "TrajGPT: Irregular Time-Series Representation Learning of Health Trajectory," IEEE Journal of Biomedical and Health Informatics, vol. 30, no. 5, pp. 3888–3900, May 2026, doi: 10.1109/JBHI.2025.3620205.

[44] A. Gayen, S. Chakraborty, J. Chakraborty, and A. Jana, "Trust@Health: A Trust-Based Multilayered Network for Scalable Healthcare Service Management," IEEE Access, vol. 13, pp. 162340–162356, 2025, doi: 10.1109/ACCESS.2025.3613326.

[45] B. Behara, M. Delrobaei, and N. Afraz, "Trusted Blockchain-Based Clinical Decision and Medication Management System for Movement Disorders," IEEE Access, vol. 13, pp. 45120–45136, 2025, doi: 10.1109/ACCESS.2025.3541289.

[46] Y. J. Msosa, A. Grauslys, Y. Zhou, T. Wang, I. Buchan, P. Langan, S. Foster, M. Walker, M. Pearson, A. Folarin, A. Roberts, S. Maskell, R. Dobson, C. Kullu, and D. Kehoe, "Trustworthy Data and AI Environments for Clinical Prediction: Application to Crisis-Risk in People With Depression," IEEE Journal of Biomedical and Health Informatics, vol. 27, no. 11, pp. 5588–5598, Nov. 2023, doi: 10.1109/JBHI.2023.3312011.